

VARUN NAUDIYAL

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PROFILE

My research interests lie in the field of Compact Object Astrophysics where I study Neutron Stars and Hybrid Stars. I aim to use multi-messenger observations of Gravitational Waves and Electromagnetic Signals in order to constrain theoretical models of the Nuclear Equation of State. I am interested in developing computational simulations to study astrophysical systems, with a particular focus on neutron-star merger remnants. I am exploring these systems through both dynamical time-evolution and the computation of quasi-static equilibrium sequences, with the aim of connecting macroscopic remnant properties to the underlying microphysics and, ultimately, to observable multi-messenger signals.

My broader interests include studying the evolution of binary star systems in the common-envelope phase. Such systems are thought to play an important channel in the formation of Binary Neutron Stars. I am also interested in accretion disk physics and protostellar evolution and hope to explore these areas further.

EDUCATION

BS-MS in Physics

Dec 2022 – Present

Indian Institute of Science Education and Research, Bhopal

Relevant Coursework:

(PHY303/304) Quantum Mechanics I & II	(PHY404) Nuclear & Particle Physics
(PHY407) Electrodynamics	(PHY615) Quantum Field Theory I*
(PHY616) General Relativity	(PHY630) Cosmology I
(PHY633) Introduction to Astrophysics	(PHY634) Advanced Statistical Mechanics

*ongoing

TECHNICAL SKILLS

Languages	Python, Fortran90, Mathematica, L ^A T _E X
Python Libraries	NumPy, SciPy, Matplotlib, Pandas
Codes	RNS (Rotating Neutron Stars), GR1D (GR 1D Hydrodynamics), LORENE
Numerical Methods	ODE integration (shooting methods, Runge–Kutta), root-finding, finite differencing

PROJECTS

Nuclear Equation of State: RMF/QHD Implementation in Neutron Stars (Ongoing) August 2026 – Present

- Studying the theoretical foundations of Quantum Hadrodynamics and Relativistic Mean Field theory as a framework for describing dense nuclear matter beyond saturation density.
- Implementing a beta-equilibrated nuclear matter EOS incorporating the baryon octet coupled to σ , ω , and ρ mesons, with leptons (electrons and muons) enforcing charge neutrality, working toward a self-consistent mean-field numerical solver.
- Actively extending the framework toward hyperon-rich matter and hybrid EOS construction incorporating a hadron-quark phase transition, with focus on the observational consequences for maximum mass and compactness.

Rotational Energy Budget of Supramassive Neutron Star Merger Remnants June 2026 – July 2026
Supervised by Dr. Ritam Mallick, IISER-B *Summer Project*

- Studied the theory of rotating neutron stars in general relativity and the turning point theorem, which identifies the onset of secular instability along constant-baryonic-mass sequences as the criterion for spin-down collapse of supramassive configurations.
- Used the RNS code to construct sequences of rotating stellar configurations for supramassive neutron stars, parameterised by baryonic mass, spanning from the mass-shedding limit down to the turning point of instability.

- Computed the extractable rotational kinetic energy along these sequences – the energy released as a supramassive remnant spins down toward collapse – and investigated to what extent an associated kilonova transient could be “super-charged” by this energy.
- Compared the computed energy budget against the observed kilonova energy of GW170817, finding that if the post-merger remnant was supramassive, it must have been close to the supramassive limit.

Exploring Neutron Stars in R^2 Gravity

June 2025 – July 2025

Supervised by Mr. Asim Kumar Saha and Dr. Ritam Mallick, IISER-B

Summer Project

- Studied $f(R)$ theories of gravity as extensions of General Relativity; derived the modified gravitational field equations for the Starobinsky R^2 model from the action principle, both by hand and via a generalised Mathematica pipeline.
- Extended the Tolman–Oppenheimer–Volkoff stellar structure equations to $f(R)$ gravity, deriving the modified pressure, mass, and scalaron evolution equations governing hydrostatic equilibrium in a curved-spacetime framework.
- Numerically integrated the modified stellar structure equations for a polytropic equation of state using a shooting method, mapping out mass-radius configurations under R^2 gravity.
- Encountered and diagnosed the well-known scalaron fine-tuning problem: the exterior solution is exponentially sensitive to the central value of the Ricci scalar, requiring precise initial conditions to produce physically regular solutions – a fundamental numerical pathology of the $f(R)$ stellar structure problem.

Numerical Relativity in 1 Dimension (Completed)

August 2025 – November 2025

Supervised by Mr. Hrishikesh Borthakur and Dr. Ritam Mallick, IISER-B

Semester Project

- Worked through the opening four chapters of Baumgarte and Shapiro’s *Numerical Relativity*, covering the 3+1 decomposition of Einstein’s equations, conformal reformulations, and the initial value problem.
- Supplemented reading with hands-on simulation: used the GR1D code to evolve a spherically symmetric polytropic star ($K = 100$, $\Gamma = 2$), verifying equilibrium stability and developing familiarity with a production-level GR hydrodynamics framework.

Self-Directed Study: X-Ray Binaries and Accretion Physics (Ongoing)

August 2026 – Present

- Surveying the observational landscape of X-ray binary systems through current review literature, developing working knowledge of accretion-driven phenomena in neutron star and black hole binaries.
- Reading *Accretion Power in Astrophysics* (Frank, King & Raine) for the theoretical foundations of accretion disk structure, angular momentum transport, and radiative processes.

POSITIONS OF RESPONSIBILITY

Core Committee Member

Aug 2024 – Mar 2025

IISER-Bhopal Astronomy Club (IBAC)

- Designed the cover page and contributed to multiple sections of the IBAC Summer Magazine.
- Co-organised IBAC-101, a club-led public lecture series introducing astronomy to a general audience; delivered a talk on star formation and protostellar evolution.
- Conducted public telescope viewing sessions using Dobsonian reflectors, handling equipment setup, alignment, and on-the-spot explanations for a non-specialist audience.
- Co-organised *Hunting the Heavens*, a riddle-and-puzzle event requiring participants to decode astronomy-based clues in order to identify objects in the night sky.

Core Committee Member, Design & Content

Aug 2024 – Nov 2024

Armageddon — Annual Tech-Fest, IISER Bhopal

- Led the Design and Content Creation team, overseeing the complete creative direction of the fest from conceptualisation through execution.
- Personally designed promotional and event materials including posters, banners, section headers, and the official fest logo.

- Authored scripts and speeches for event hosts, adapting tone and content across different segments of the programme.

Peer Counselor

Aug 2023 – Mar 2024

IISER-Bhopal Student Wellness & Counselling Team

- Provided confidential peer support to fellow students navigating academic stress, personal challenges, and adjustment difficulties, maintaining a non-judgmental and empathetic approach.
- Collaborated with the wider counselling team to organise outreach activities aimed at reducing stigma around mental health within the student community.